Fiscal Year:	FY 2007	**Task Last Updated:**	FY 03/06/2008
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Project Title:	FUNDAMENTAL EVALUATION OF WATER MIST FOR FIRE SAFETY IN SPACECRAFT AND EXTRATERRESTRIAL ENVIRONMENTS		
Division Name:	Physical Sciences		
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Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
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Task Description:	This investigation provides an integrated suite of tasks that involve both experimental studies and detailed modeling of fine water mist in fire suppression scenarios relevant to spaceflight needs. The overall goal of this fundamental study and the resulting numerical code, validated with experiments, is to evaluate the role of water mist in the next generation of fire safety systems on spacecraft and habitation modules on the surface of other planets. Ultra-fine water mist driven by air or by nitrogen is compared to carbon dioxide and nitrogen to evaluate their performance to suppress and extinguish both direct and highly obstructed fires involving energized wires, electrical components on circuit boards, and cloth. Extinction time and suppressant amount are used as a measure of agent effectiveness. The numerical-experimental study is based on a hierarchical model. First, a study of simplified scenarios is performed at a detailed level. Second, the critical elements of the detailed model are extracted and then encapsulated in a reduced-order manner for inclusion in the next higher-level scenario. The process is continued until a robust understanding of fire suppression at the level of a Space Shuttle Mid-Deck Locker is obtained. The experiments are performed in normal gravity for an accurate validation of the numerical model and to provide observations of fire suppression behavior under realistic spacecraft fire scenarios.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results on the effectiveness of the various suppressant agents used in this project serve to determine an optimum fire-fighting technique for space and for Earth applications. While the main emphasis is on fighting fires on spacecraft, the results are generalized for all gravity conditions, including Earth's gravity. The software code developed serves as a tool for designing fire suppression systems for aircraft, ships, museums, libraries, and a variety of enclosed spaces.
Task Progress:	An experimental investigation on the effectiveness of various suppressant agents in spacecraft fires under normal-gravity conditions has been conducted. Ultra-fine water mist clouds (mean diameter of 8 microns) driven by air or by nitrogen (N2) are compared to single-fluid agents such as carbon dioxide (CO2) and pure N2 to evaluate their performance to suppress and extinguish typical fires that may be encountered in spacecraft such as energized wires, electrical components on circuit boards, and cloth. The suppressants are injected into a chamber with similar dimensions to the Space Shuttle Mid-deck Locker at low (0.2 g/s), medium (1.0 g/s) and high (2.0 g/s) mass-flow rates. A highly obstructed path to the fire is provided by a large baffle located perpendicular to the agent stream, between the nozzle and the fire. Extinction time and amount of suppressant agent needed to put out the flames are measured and used in the comparative study, along with other performance variables related to practical aspects of fighting fires under spacecraft conditions. a) Fire Scenario 1: Overheated Wire. As expected for agents with similar heat capacities per unit mass, the extinction time for CO2 and N2 is quite comparable. Interestingly, ultra-fine mist driven by air is not as effective as the gaseous agents for this fire scenario, taking considerably longer to extinguish the fire. This apparent weakness is in sharp contrast to the advantages of using water mist in the burning-element and burning-cloth cases. This discrepancy may be explained by a combination of two factors: a) the overwhelming heat generated by the electrical power applied to the wire (35 W), approximately half of the fire power, and b) the “small-fire syndrome,” a phenomenon frequently observed in weekly burning fires suppressed with water, where the small fire is able to survive with barely adequate levels of oxygen by reducing the water evaporation rate below a critical value. It is believed that these two reasons may explain why the water droplets are not being efficiently used to put out the fire. During the experiments, it is observed that the mist quickly reaches the fire after less than 5 seconds from injection and immediately suppresses it by significantly diminishing its size and its spread rate. Nevertheless, after that rapid initial suppression, some amount of water mist appears to be spent cooling the hot copper wire by evaporating around it, another group of droplets just re-circulates inside the test chamber, while the rest of the droplets concentrates near the small fire. To verify this hypothesis, the ultra-fine mist is introduced at the slow mass-flow rate of 0.2 g/s, resulting in 65% less water needed to put out the fire even though it takes much longer to extinguish it, indicating that at higher mass-flow rates, the water is not effectively used to control the fire (on a mass basis). If extinction time is the main design parameter, it is interesting to note that although it takes more than three times longer to extinguish the fire, the final position of the flame at the low mass-flow rate case is just a few millimeters past the spot where the flame extinguishes under the medium mass-flow rate case. Obviously, ultra-fine mist acts as an extremely efficient suppressant, but it has a more difficult time to completely extinguish the small remaining flame, as compared to the gaseous agents. To verify the “small-fire syndrome” hypothesis, 0.5 g/s of ultra-fine mist are introduced in the chamber driven by 0.5 g/s of N2. This combination effectively cools down the wire, rapidly suppresses the spread rate of the flame, and quickly puts out the remaining small fire. For the water-mist/N2 combination, both the extinction time and the total amount of suppressant used to extinguish the flame are comparable to the values obtained for the gaseous agents. It is important to point out that neither CO2 nor N2 are capable of suppressing the fire at mass-flow rates lower than 1 g/s. Under these conditions, most of the time the wire breaks before the flame is extinguished, and on other occasions the insulation completely melts down and falls to the bottom of the chamber, making it impossible to quantify the effectiveness of the agent. For this particular case, the unique suppression properties of water mist in conjunction with the oxygen-depletion attributes of N2 result in a superior combination with excellent performance. b) Fire Scenario 2: Burning Element on a Circuit Board. In the simulated case of a burning element on a circuit board case, only 25 W are applied to the heater cartridge to preheat the PMMA rod, allowing the water droplets to concentrate on putting out the fire instead of cooling the heated element. Consequently, ultra-fine mist gives the best results based on time and suppressant amount, followed closely by the two other gaseous agents. As with the overheated wire case, the ultra-fine mist is also introduced at the slow mass-flow rate of 0.2 g/s, resulting in less water needed to put out the fire even though it takes much longer to extinguish it. In addition, although it takes more than four times as long to extinguish the fire, the final position of the flame at the low mass-flow rate case is just a few millimeters past the spot where the flame extinguishes with the medium mass-flow rate. This is another case where fire suppression is effectively and quickly accomplished by the ultra-fine mist with the rest of the time spent to completely extinguish the small remaining flame. The gaseous agents are again incapable of putting the fire out at the low mass-flow rate with the flame completely consuming all the PMMA material. c) Fire Scenario 3: Burning Cloth. For the burning cloth case, a representative example of a fire burning completely unaided by any external heating, ultra-fine mist clearly is the most effective suppressant as compared to the CO2 and N2 gases. On a mass basis, its

effectiveness its further enhanced when ultra-fine mist is introduced at the low mass-flow rate of 0.2 g/s, while the gaseous agents are incapable of putting the fire out at this lower rate. For the high mass-flow rate case, CO2 and N2 not only introduce convective currents that would enhance the flame strength under low-gravity conditions, but they also blow away the smoldering embers that remain in the post-flame zone on top of the burning rag. An added benefit of using water droplets in this particular case is the ability of the water to extinguish the smoldering embers considerably faster than the other two agents. This is due to the moisture absorbed by the cloth that forms an obstructive barrier for further flame propagation.

Thus, on a mass-basis comparison, ultra-fine mist is found to be more effective than CO2 and N2, with the exception of the burning-wire scenario with a high preheating power. In this case, ultra-fine mist is used to not only extinguish the fire, but also to cool the hot copper wire. This fact, in conjunction with the “small-fire syndrome” characteristic of water-mist suppressed fires, results in longer extinction times for ultra-fine mist, as compared to gaseous agents. Although at first the longer extinction times and additional mass of agent may appear as a disadvantage of water mist against CO2 or N2, the cooling effect of the water mist has the advantage of preventing the rupture of the wire, as it occurs when CO2 and N2 are applied at low mass-flow rates. This cooling effect may be essential to avoid disrupting the operation of critical equipment onboard the spacecraft while the fire is being suppressed. If shorter extinction times are desired, ultra-fine mist driven by N2 can be used to combine the unique suppression properties of water mist in conjunction with the oxygen-depletion attributes of N2. It is observed that the use of this mixed-agent suppressant results in a superior combination with excellent performance. These observations are supported and explained by the numerical model results which compare favorably to the observed experimental data and emphasize the advantage of using a combined water-mist/nitrogen system to effectively cool, suppress, and extinguish this type of small fires. On a parallel study of the behavior of ultra fine mist as it approaches the fire, the numerical analysis of the idealized behavior of a mist parcel with droplet sizes on the order of a few microns shows that collisions have negligible influence on the droplet size distribution evolution over time scales commensurate with the flow residence time. In addition, it is found that drag, heat transfer, and evaporation occur on time scales that are significantly shorter than that of the flow field, indicating that droplets will trace the flow of the continuous phase, and quickly reach thermodynamic equilibrium with it.

Ultra-fine mist is also effective on reducing smoldering extinction times with cloth fires by adding moisture to the burning rag and preventing the blowing of hot smoldering embers that can cause ignition of neighboring material in a tightly packed spacecraft under low-gravity conditions. In addition, the low-momentum injection of water mist may prevent the strengthening of the weak-burning low-gravity fires by reducing the convective currents, which are observed with the gaseous agents injected at high flow rates. Finally, the low momentum and pseudo-gas properties of ultra-fine mist with lower-than-10-micron droplets are also more effective than the high-momentum and larger droplet sizes generated by high-pressure water-mist sprays, which make water droplets incapable of moving around obstacles and reaching the fire.

These results point to ultra-fine water mist (or a combination of water mist driven by N2) as a promising candidate to be used as agent in the next-generation spacecraft fire suppression system. The final selection of the optimum suppressant agent will depend on a detailed assessment of all the relevant design parameters and their relative importance for this unique application.

Bibliography Type:	Description: (Last Updated: 03/21/2008)
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